

CHAPTER 4

DATA LINK LAYER

Chapter Summary

The physical layer (also called layer 1) is the physical connection between the computers and/or devices in the network. This chapter examines how the physical layer operates. It describes the most commonly used media for network circuits and explains the basic technical concepts of how data is actually transmitted through the media. Three different types of transmission are described: digital transmission of digital computer data; analog transmission of digital computer data; and digital transmission of analog voice data. You do not need an engineering-level understanding of the topics to be an effective user and manager of data communication applications. It is important, however, that you understand the basic concepts, so this chapter is somewhat technical.

Learning Objectives

After reading this chapter, students should be able to:

- understand the role of the data link layer
- become familiar with two basic approaches to controlling access to the media
- become familiar with common sources of error and their prevention
- understand three common error detection and correction methods
- become familiar with several commonly used data link protocols

Key Terms

access request	flow control	negative acknowledgment
acknowledgment	forward error correction	(NAK)
(ACK)	frame	odd parity
amplifier	Gaussian noise	overhead bits
asynchronous transmission	Go-Back-N ARQ	parity bit
attenuation	Hamming code	parity check
Automatic Repeat reQuest	high-level data link control	Point-to-Point Protocol
(ARQ)	(HDLC)	(PPP)
burst error	hub polling	polling
checksum	impulse noise	repeater
contention	information bits	roll-call polling
continuous ARQ	intermodulation noise	sliding window
controlled access	line noise	start bit
cross-talk	Link Access	stop-and-wait ARQ
cyclical redundancy check	Protocol-Balanced (LAP-	stop bit
(CRC)	B)	synchronization
echo	Link Access Protocol for	synchronous transmission
efficiency	Modems (LAP-M)	throughput
error detection	logical link control (LLC)	token passing
error prevention	sublayer	transmission efficiency
error rate	media access control	white noise
Ethernet (IEEE 802.3)	media access control	
even parity	(MAC) sublayer	

Chapter Outline

1. INTRODUCTION
2. MEDIA ACCESS CONTROL
 - a. Contention
3. Controlled Access
 - a. Relative Performance
4. ERROR CONTROL
 - a. Sources of Error
 - b. Error Prevention
 - c. Error Detection
 - d. Error Correction via Retransmission
 - e. Forward Error Correction
 - f. Error Control in Practice
5. DATA LINK PROTOCOLS
 - a. Asynchronous Transmission
 - b. Synchronous Transmission
6. TRANSMISSION EFFICIENCY
7. IMPLICATIONS FOR MANAGEMENT
- SUMMARY

Answers to Textbook Exercises

1. What does the data link layer do?

The data link layer controls the way messages are sent on the physical media. The data link layer handles three functions: media access control, message delineation, and error control. The data link layer accepts messages from the network layer and controls the hardware that actually transmits them. The data link layer is responsible for getting a message from one computer to another without errors. The data link layer also accepts streams of bits from the physical layer and organizes them into coherent messages that it passes to the network layer.

2. What is media access control, and why is it important?

Media access control handles when the message gets sent. Media access control becomes important when several computers share the same communication circuit, such as a point-to-point configuration with a half duplex line that requires computers to take turns, or a multipoint configuration in which several computers share the same circuit. Here, it is critical to ensure that no two computers attempt to transmit data at the same time -- or if they do, there must be a way to recover from the problem. Media access control is critical in local area networks.

3. Under what conditions is media access control unimportant?

With point-to-point full duplex configurations, media access control is unnecessary because there are only two computers on the circuit and full duplex permits either computer to transmit at anytime. There is no media access control.

4. Compare and contrast roll-call polling, hub polling (or token passing), and contention.

With roll call polling, the front end processor works consecutively through a list of clients, first polling terminal 1, then terminal 2, and so on, until all are polled. Roll call polling can be modified to select clients in priority so that some get polled more often than others. For example, one could increase the priority of terminal 1 by using a polling sequence such as 1, 2, 3, 1, 4, 5, 1, 6, 7, 1, 8, 9.

Hub polling is often used in LAN multipoint configurations (i.e., token ring) that do not have a central host computer. One computer starts the poll and passes it to the next computer on the multipoint circuit, which sends its message and passes the poll to the next. That computer then passes the poll to the next, and so on, until it reaches the first computer, which restarts the process again.

Contention is the opposite of controlled access. Computers wait until the circuit is free (i.e., no other computers are transmitting), and then transmit whenever they have data to send. Contention is commonly used in Ethernet local area networks.

5. Which is better, controlled access or contention? Explain.

The key consideration for which is better is throughput -- which approach will permit the largest amount of user data to be transmitted through the network. In most of the 1990s, contention approaches worked better than controlled approaches for small networks that have low usage. In this case, each computer can transmit when necessary, without waiting for permission. In high volume networks, where many computers want to transmit at the same time, the well-controlled circuit originally prevented collisions and delivered better throughput in such networks. Today contention-based systems have been improved to the point where they deliver substantially better throughput and are competitive because of hardware cost considerations.

6. Define two fundamental types of errors.

There are two fundamental types of errors: human errors and network errors. Human errors, such as a mistake in typing a number, usually are controlled through the application program. Network errors, such as those that occur during transmission, are controlled by the network hardware and software. There are two categories of network errors: corrupted data (data that have been changed) and lost data.

7. Errors normally appear in _____, which is when more than one data bit is changed by the error-causing condition.

Errors normally appear in bursts, which is when more than one data bit is changed by the error-causing condition.

8. Is there any difference in the error rates of lower-speed lines and of higher-speed lines?

Yes, normally lower speed lines have higher error rates because (1) leased lines can be conditioned to prevent noise, but dial-up lines can not and (2) dial-up lines have less stable transmission parameters.

9. Briefly define *noise*.

Noise consists of undesirable electrical signals, or, in the instance of fiber optic cable, undesirable light. Noise is typically introduced by equipment or natural disturbances, and it can seriously degrade the performance of a communication circuit. Noise manifests itself as extra bits, missing bits, or bits that have been "flipped," (i.e., changed from 1 to 0 or vice versa).

10. Describe four types of noise. Which is likely to pose the greatest problem to network managers?

The following list summarizes the major sources of error. The first six are the most important; the last three are more common in analog rather than digital circuits.

Line outages are a catastrophic cause of errors and incomplete transmission. Occasionally, a communication circuit fails for a brief period. This type of failure may be caused by faulty telephone end office equipment, storms, loss of the carrier signal, and any other failure that causes a short circuit. When constructing and designing redundant networks that are fault survivable, this is usually called designing for the “farmer with a back hoe” problem.

White noise or gaussian noise (the familiar background hiss or static on radios and telephones) is caused by the thermal agitation of electrons and therefore is inescapable. Even if the equipment was perfect and the wires were perfectly insulated from any and all external interference, there still would be some white noise. White noise usually is not a problem unless it becomes so strong that it obliterates the transmission. In this case, the strength of the electrical signal is increased so it overpowers the white noise; in technical terms, we increase the signal to noise ratio.

Impulse noise (sometimes called spikes) is the primary source of errors in data communications. Some of the sources of impulse noise are voltage changes in adjacent lines, lightning flashes during thunderstorms, fluorescent lights, and poor connections in circuits.

Cross-talk occurs when one circuit picks up signals in another. It occurs between pairs of wires that are carrying separate signals, in multiplexed links carrying many discrete signals, or in microwave links in which one antenna picks up a minute reflection from another antenna. Cross-talk between lines increases with increased communication distance, increased proximity of the two wires, increased signal strength, and higher frequency signals. Wet or damp weather can also increase cross-talk. Like white noise, cross-talk has such a low signal strength that it normally is not bothersome.

Echoes can cause errors. Echoes are caused by poor connections that cause the signal to reflect back to the transmitting equipment. If the strength of the echo is strong enough to be detected, it causes errors. Echoes, like cross-talk and white noise, have such a low signal strength that they normally are not bothersome. In networks, echo suppressors are devices that reduce the potential for this type of error. Echoes can also occur in fiber optic cables when connections between cables are not properly aligned.

Attenuation is the loss of power a signal suffers as it travels from the transmitting computer to the receiving computer. Some power is absorbed by the medium or is lost before it reaches the receiver. This power loss is a function of the transmission method and circuit medium. High frequencies lose power more rapidly than low frequencies during transmission, so the received signal can thus be distorted by unequal loss of its component frequencies. Attenuation increases as frequency increases or as the diameter of the wire decreases, or as the distance of the transmission increases. Repeaters can be used in a digital environment to correct for attenuation due to distance, where amplifiers can be used to boost diminishing or attenuating analog signals over longer distances. A repeater will perfectly replicate the incoming, distorted digital signal and send it on deeper into the network as if new. An amplifier will boost an attenuating analog signal, but also boost the error noise in the signal as it does so. Fewer repeaters are necessary as compared to amplifiers to correct for attenuation, thus

helping to make digital more cost effective when compared to analog transmission in controlling for noise.

Intermodulation noise is a special type of cross-talk. The signals from two circuits combine to form a new signal that falls into a frequency band reserved for another signal. On a multiplexed line, many different signals are amplified together, and slight variations in the adjustment of the equipment can cause intermodulation noise. A maladjusted modem may transmit a strong frequency tone when not transmitting data, thus producing this type of noise.

Jitter may affect the accuracy of the data being transmitted because minute variations in amplitude, phase, and frequency always occur. The generation of a pure carrier signal in an analog circuit is impossible. The signal may be impaired by continuous and rapid gain and/or phase changes. This jitter may be random or periodic.

Harmonic distortion usually is caused by an amplifier on a circuit that does not correctly represent its output with what was delivered to it on the input side.

Phase hits are short-term shifts "out of phase," with the possibility of a shift back into phase.

11. How do amplifiers differ from repeaters?

An amplifier takes the incoming signal, increases its strength, and retransmits it on the next section of the circuit. They are typically used on analog circuits such as the telephone company's voice circuits. On analog circuits, it is important to recognize that the noise and distortion are also amplified, along with the signal.

Repeaters are commonly used on digital circuits. A repeater receives the incoming signal, translates it into a digital message, and retransmits the message. Because the message is re-created at each repeater, noise and distortion from the previous circuit are not amplified.

12. What are three ways of reducing errors and the types of noise they affect?

Shielding (protecting wires by covering them with an insulating coating) is one of the best ways to prevent impulse noise, cross-talk and intermodulation noise.

Moving cables away from sources of noise (especially power sources) can also reduce impulse noise cross-talk and intermodulation noise. For impulse noise, this means avoiding lights and heavy machinery. Locating communication cables away from power cables is always a good idea. For cross-talk, this means physically separating the cables from other communication cables.

Cross-talk and intermodulation noise is often caused by improper multiplexing. Changing multiplexing techniques (e.g., from FDM to TDM), or changing the frequencies or size of the guardbands in frequency division multiplexing can help.

Many types of noise (e.g., echoes, white noise, jitter, harmonic distortion) can be caused by poorly maintained equipment or poor connections and splices among cables. The solution here is obvious: tune the transmission equipment and redo the connections.

To avoid attenuation, telephone circuits have repeaters or amplifiers spaced throughout their length.

13. Describe three approaches to detecting errors, including how they work, the probability of detecting an error, and any other benefits or limitations.

Three common error detection methods are parity checking, longitudinal redundancy checking, and polynomial checking (particularly checksum and cyclic redundancy checking).

One of the oldest and simplest error detection methods is parity. With this technique, one additional bit is added to each byte in the message. The value of this additional parity bit is based on the number of 1's in each byte transmitted. This parity bit is set to make the total number of ones in the byte (including the parity bit) either an even number or an odd number. Any single error (a switch of a 1 to a 0 or vice versa) will be detected by parity, but it cannot determine which bit was in error. But, if two bits are switched, the parity check will not detect any error. Parity can detect errors only when an odd number of bits have been switched; any even number of errors cancel each other out. Therefore, the probability of detecting an error, given that one has occurred, is only about 50 percent. Many networks today do not use parity because of its low error detection rate.

Polynomial checking adds a character or series of characters to the end of the message based on a mathematical algorithm. With the checksum technique, a checksum (typically one byte) is added to the end of the message. The checksum is calculated by adding the decimal value of each character in the message, dividing the sum by 255, and using the remainder as the checksum. The receiver calculates its own checksum in the same way and compares it with the transmitted checksum. If the two values are equal, the message is presumed to contain no errors. Use of checksum detects close to 95 percent of the errors for multiple bit burst errors. The most popular polynomial error checking scheme is cyclical redundancy check (see the answer to # 16 below for more discussion). The probability of detecting an error is nearly 100% or, in some cases, 100%.

14. Briefly describe how even parity and odd parity work.

Even parity is when the seven bits of an ASCII character have an even (2, 4, or 6) number of 1s, and therefore a 0 is placed in the eighth parity position. Odd parity is when the seven bits of an ASCII character have an odd (1, 3, 5, or 7) number of 1s, and therefore a 1 is placed in the eighth parity position.

15. Briefly describe how checksum works.

Checksum error checking adds a checksum (typically 1 byte) is added to the end of the message. The checksum is calculated by adding the decimal value of each character in the message, dividing the sum by 255, and then using the remainder as the checksum. The same approach is used at the receiving end. If the receiver gets the same result, the block has been received correctly.

16. How does cyclical redundancy checking (CRC) work?

Cyclical redundancy check (CRC) adds 8, 16, 24 or 32 bits to the message. With CRC, a message is treated as one long binary number, P . Before transmission, the data link layer (or hardware device) divides P by a fixed binary number, G , resulting in a whole number, Q , and a remainder, R/G . So, $P/G = Q + R/G$. For example, if $P = 58$ and $G = 8$, then $Q = 7$ and $R = 2$. G is chosen so that the remainder R will be either 8 bits, 16 bits, 24 bits, or 32 bits.

The remainder, R , is appended to the message as the error checking characters before transmission. The receiving hardware divides the received message by the same G , which generates an R . The receiving hardware checks to ascertain whether the received R agrees with the locally generated R . If it does not, the message is assumed to be in error.

17. How does forward error correction work? How is it different from other error-correction methods?

Forward error correction uses codes containing sufficient redundancy to prevent errors by detecting and correcting them at the receiving end without retransmission of the original message. The redundancy, or extra bits required, varies with different schemes. It ranges from a small percentage of extra bits to 100 percent redundancy, with the number of error detecting bits roughly equaling the number of data bits.

18. Under what circumstances is forward error-correction desirable?

Forward error correction is commonly used in satellite transmission. A round trip from the Earth station to the satellite and back includes a significant delay. Error rates can fluctuate depending on the condition of equipment, sun spots, or the weather. Indeed, some weather conditions make it impossible to transmit without some errors, making forward error correction essential. Compared to satellite equipment costs, the additional cost of forward error correction is insignificant.

19. Compare and contrast stop-and-wait ARQ and continuous ARQ.

With stop-and-wait ARQ, the sender stops and waits for a response from the receiver after each message or data packet. After receiving a packet, the receiver sends either an acknowledgment (ACK) if the message was received without error, or a negative acknowledgment (NAK) if the message contained an error. If it is a NAK, the sender resends the previous message. If it is an ACK, the sender continues with the next message. Stop-and-wait ARQ is by definition, a half duplex transmission technique.

With continuous ARQ, the sender does not wait for an acknowledgment after sending a message; it immediately sends the next one. While the messages are being transmitted, the sender examines the stream of returning acknowledgments. If it receives an NAK, the sender retransmits the needed messages. Continuous ARQ is by definition a full duplex transmission technique, because both the sender and the receiver are transmitting simultaneously (the sender is sending messages, and the receiver is sending ACKs and NAKs).

20. Which is the simplest (least sophisticated) protocol described in this chapter?

An argument could be made for SDLC, HDLC, or PPP. Each of these are similar in many ways.

21. Describe the frame layouts for SDLC, Ethernet, and PPP.

Each SDLC frame begins and ends with a special bit pattern, known as the flag. The address field identifies the destination. The length of the address field is usually 8 bits but can be set at 16 bits; all computers on the same network must use the same length. The control field identifies the kind of frame that is being transmitted, either information or supervisory. An information frame is used for the transfer and reception of messages, frame numbering of contiguous frames, and the like. A supervisory frame is used to transmit acknowledgments (ACKs and NAKs). The message field is of variable length and is the user's message. The frame check sequence field is a 16-bit or 32-bit cyclical redundancy checking (CRC) code.

For a typical Ethernet packet, the destination address specifies the receiver, while the source address specifies the sender. The length indicates the length in 8-bit bytes of the message portion of the packet. The LLC control and SNAP control are used to pass control information between the sender and receiver. These are often used to indicate the type of network layer protocol the packet contains (e.g., TCP/IP or IPX/SPX as described in Chapter 6). The maximum length of the message is 1492 bytes. The packet ends with a CRC-32 frame check sequence used for error detection.

The PPP frame is similar to the SDLC frame. The frame starts with a flag and has a one-byte address. It also contains a control field which is rarely used. The protocol field indicates what type of data is contained. The message portion is variable in length and may be up to 1,500 bytes long. The frame check sequence is either CRC-16 or -32. The frame ends with a flag.

22. What is transmission efficiency?

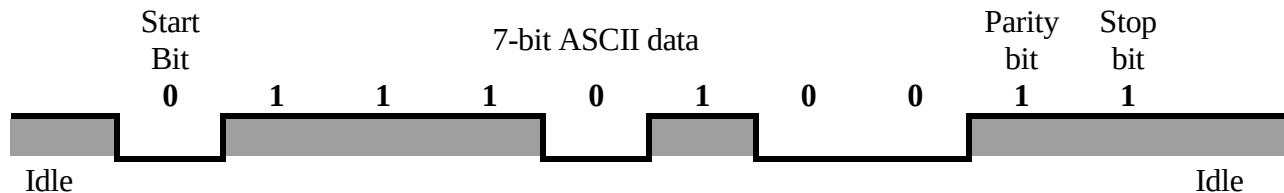
Transmission efficiency is defined as the total number of information bits (i.e., bits in the message sent by the user) divided by the total bits in transmission (i.e., information bits plus overhead bits).

23. How do information bits differ from overhead bits?

Information bits are those used to convey the user's meaning. Overhead bits are used for purposes such as error checking, and marking the start and end of characters and packets.

24. Are stop bits necessary in asynchronous transmission? Explain using a diagram.

Stop bits in asynchronous transmission are necessary to return the state of the transmission medium to the idle state (such as +3v in Figure 4-9). The signal must return to the idle state in order for the start bit to be recognized as such.



25. During the 1990s, there was intense competition between two technologies (10 Mbps Ethernet and 16 Mbps token ring) for the LAN market. Ethernet was promoted by a consortium of vendors while token ring was primarily an IBM product, even though it was standardized. Ethernet won, and no one talks about token ring anymore. Token ring used a hub-polling-based approach. Outline a number of reasons why Ethernet might have won. Hint: the reasons were both technical and business.

26. Under what conditions does a data link layer protocol need an address?

At some point in the networking process, the device's data link layer must be addressed by the layer 3 protocol, no matter what the particular data link layer protocol that is running on the network.

27. Are large frame sizes better than small frame sizes? Explain.

Selecting the right frame size can have a great effect on performance. There is an optimal frame size that is not so small that packets have low efficiency by carrying too little information for too much overhead, nor so large as to incur the risk of more errors and thus longer and more frequent retransmission. The optimal frame size is dependent on the specific application and the pattern of messages it generates.

28. What media access control technique does your class use?

Each class will have a different answer.

29. Show how the word "HI" would be sent using asynchronous transmission using even parity (make assumptions about the bit patterns needed). Show how it would be sent using Ethernet.

H → 0100 1000

I → 0100 1001

Adding single bit, even parity:

H → 0100 1000 0

I → 0100 1001 1

Thus, asynchronous with 0 start bit, 1 stop bit:

0

0 0100 1000 0 1
0 0100 1001 1 1

Ethernet packet would have frame with data inserted within it as follows:

7 byte preamble

1 byte frame start

6 bytes destination

6 bytes source

4 bytes VLAN (optional)

2 bytes length

1 byte control information (DSAP)

1 byte control information (SSAP)

1-2 bytes control

Data: H → 0100 1000

I → 0100 1001

Frame check sequence

Mini-Cases

I. Smith, Smith, Smith, and Smith

Smith, Smith, Smith, and Smith is a regional accounting firm that is building a new headquarters building. The building will have a backbone network that connects eight LANs (two on each floor). They are very concerned with network errors. What advice would you give them in the design of the building and network cable planning that would help reduce network errors?

Building and network cable planning should take into account the recommended maximum lengths for media and protocols, avoid running network cable alongside power cable, include adequate shielding in specifying cable, keep cable away from lights and heavy machinery, and include adequate repeaters or amplifiers.

II. Worldwide Charity

Worldwide Charity is a charitable organization whose mission is to improve education in developing countries. What data link layer protocols should WC staff use for file transfer? What range of frame sizes is likely to be used?

Recommended is Zmodem, because it has a powerful error detection method (CRC-32) with continuous ARQ. Note that Zmodem dynamically adjusts packet size to communication circuit conditions to increase efficiency.

III. Industrial Products

What would you suggest Clarence Hung do to investigate why the network is reported to be slow over the past week?

He should check to see if any new computers have been added to the network, which may result in collisions if the system uses a contention approach to control network access.

IV. Alpha Corp.

1. Without considering transmission efficiency, how large an Internet connection would you recommend in terms of bits per second (assuming that each byte is 8 bits in length)?

Since transmission efficiency is not a consideration, one could get by with V.92 modem at 56 Kbps; however keep in mind this would be very slow. If given the choice, avoid the potential problem, consider SDLC.

2. Assuming they use a synchronous data link layer protocol with an efficiency of about 90%, how large an Internet connection would you recommend?

You would need an Internet connection that could handle 2,000 and 4,000 bytes. This number should be sufficient; however, if you would like to fine tune the exact number of bits, a system manager can use the TRIB calculation.

TRIB = Number of information bits accepted, divided by, the total time required to get the bits accepted

3. Suppose Alpa wants to be sure that its Internet connection will provide sufficient capacity for the next 2 years, how large an Internet connection would you recommend?

Alpa is transmitting large sums of data every hour; thus, a Zmodem might be used as it can handle a message length up to 1,024 bites. Migrating to a different protocol might also prove efficient, as the general rule is that the larger the message field, the more efficient the protocol.

Next Day Air Service Case Study

1. What file transfer protocols would you recommend? Be prepared to support your recommendations.

Student answers will vary here. However, the following are some reasonable solutions.

For the synchronous modem transfer - High-level Data Link Control (HDLC), LAP-B, Token Ring (IEEE 802.5), or Ethernet (IEEE 802.3).

HDLC is a formal standard developed by the International Organization for Standardization (ISO). HDLC is essentially the same as SDLC. The primary differences are that the address and control fields can be longer than in the SDLC frame. HDLC also has several additional benefits, such as a larger sliding window. It uses a controlled access media access protocol.

Link Access Procedure-Balanced (LAP-B) uses the same frame structure as HDLC, but is a scaled down version of HDLC.

Ethernet (IEEE 802.3) - Ethernet was developed jointly by Digital, Intel, and Xerox in the early 1980's. Since then, Ethernet has become a formal standard, called IEEE 802.3. Ethernet is a byte-count protocol in that instead of using special characters or bit patterns to mark the end of a frame, it includes a field that specifies the length of the message portion of the frame. It uses a contention media access protocol. Contention access could be a problem at peak loads. However, it should be minimal with the excess bandwidth in this network.

2. Prepare a brief position paper on the types of errors you can expect in the NDAS network and the steps you believe NDAS can take to prevent, detect, and correct these errors.

There are two basic types of errors: human errors and network errors. Human errors are (1) mistakes resulting from doing a job, (2) intentional acts that are illegal or malicious by (a) employees, and (b) outsiders who gain access to the network errors, such as a mistake in keying a number. There are two categories of network errors: corrupted data (data that has been changed) and lost data.

Human errors can be reduced by: (1) careful training of NDAS personnel, (2) error checking routines in application software and well designed user interfaces, (3) management attention to data entry quality, and (4) good system security.

Network errors can occur for a variety of physical reasons. The 11 basic categories of errors on networks, their effects, and what can be done about them, are given below.

- a. White or Gaussian
 - (1) Occurs when electrons cause thermal agitation
 - (2) Manifests itself as a hiss or static
 - (3) Inescapable, regardless of protective measures
- b. Impulse noise or spikes

- (1) Occurs when there is a sudden surge of power
- (2) Primary source of data communication errors
- (3) Caused by
 - (a) Voltage changes in adjacent lines or circuitry
 - (b) Telephone switching equipment
 - (c) Arcing of relays at older telephone exchanges offices
 - (d) Tones used by network signaling
 - (e) Maintenance equipment during line testing
 - (f) Lighting and thunderstorm
 - (g) Intermittent electrical connections in the data communication equipment
- (4) Prevention or reduction
careful installation of network equipment, attention to impulse noise or spikes when locating equipment, and good equipment maintenance
- c. Cross-talk
 - (1) Occurs when a line picks up a signal from an adjacent line
 - (2) Occurs in
 - (a) Line pairs carrying separate signals
 - (b) Multiplexed links carrying many discrete (digital) signals
 - (c) Microwave links where the antenna picks up part of a signal from another antenna on the same tower
 - (d) Hardwired circuits that run parallel, are too close together, and are not balanced electrically.
 - (3) Increases with
 - (a) Increased communication distance
 - (b) Increased proximity of two wires
 - (c) Increased signal strength
 - (d) Higher frequency signals
 - (4) Prevention or reduction
Careful installation of network equipment and cabling, good cable shielding
- d. Echoes and echo suppressors
 - (1) Caused by echo suppressors changing balance in the line, making the signal reflect back down the line at reduced strength
 - (2) Normally echoes are low signal strength, but cause errors when higher
 - (3) Prevention or reduction
Use leased lines
- e. Intermodulation noise
 - (1) A specialized type of cross-talk
 - (2) Occurs when signals from two independent lines intermodulate and form a new signal that falls into a frequency band differing from both inputs and usually one reserved for another signal.
 - (3) Caused by
 - (a) Slight maladjustment of multiplex equipment
 - (b) Poorly adjusted modem that transmits a frequency tone when not transmitting data
 - (4) Prevention or reduction
Good equipment maintenance

- f. Amplitude noise
 - (1) Occurs when the power load changes suddenly
 - (2) Caused by
 - (a) Faulty amplifiers
 - (b) Dirty contacts with variable resistance
 - (c) Sudden loads when new circuits are switched on during busy times
 - (d) Maintenance work
 - (e) Switching to a different transmission line
 - (3) Prevention or reduction
 - Good equipment maintenance
- g. Line outages
 - (1) Catastrophic cause of errors and incomplete transmissions
 - (2) Occurs when transmission power stops
 - (3) Caused by
 - (a) Faulty office exchange equipment
 - (b) Storms
 - (c) Loss of the carrier signal
 - (d) Brownouts by power companies during high usage periods
 - (e) Other failures that cause an open line or short circuit
 - (4) Prevention or reduction
 - Not much can be done, but good equipment maintenance is a help
- h. Attenuation
 - (1) Occurs when the signal loses power as it travels down a line
 - (2) Caused by the transmission medium absorbing the power, which gradually weakens the signal
 - (3) Prevented by repeaters/amplifiers that boost the signal strength before sending it on to the next repeater/amplifier
 - (4) Is a function of both the transmission method and circuit medium
 - (5) Increases as the
 - (a) Frequency increases
 - (b) Wire diameter increases
- i. Attenuation distortion
 - (1) Caused when high frequencies lose power and the received signal gets distorted by unequal loss of its component frequencies
 - (2) Prevented by equalizers that compensate for attenuation distortion
- j. Delay distortion
 - (1) Occurs when a signal is delayed more at some frequencies than at others and the bits arrive at their destination at different times
 - (2) Prevented by an equalizer that compensates for delay distortion
- k. Jitter
 - (1) Impossible to eliminate
 - (2) Caused by minute variations in amplitude, phase, or frequency
- l. Harmonic distortion
 - (1) Caused by an amplifier that does not represent the correct output with what was delivered to it as an input

- (2) Phase hits are short-term shifts out of phase with the possibility of shifting back into phase
- (3) Prevention or reduction
 - Good equipment maintenance

In summary, common sense precautions and good equipment maintenance are the best approaches for NDAS to prevent or reduce errors, since a number of the sources of network errors are beyond the control of NDAS.

Error detection and correction is the function of the data link protocol selected by NDAS. There protocols and their error detection and correction techniques are discussed in question 1. HDLC for synchronous file transfer is a good approach for NDAS.

Additional Content

Teaching Notes

I start by revisiting the network model introduced in Chapter 1 to help students put this material in context.

I spend time making sure the students understand what media access control is. This section will be revisited in detail in Chapter 6 on LANs and I want the students to have a good foundation. I just try to get the issue of controlling access to the media on the table at this point.

In my teaching on error control, my goal is to ensure that students understand the cause of errors, how to detect them (parity, LRC, CRC) and how to correct them. I do not cover forward error correction except to discuss it is possible. Likewise, I focus only on the most straightforward sources of error (white noise, impulse noise, cross-talk, echo, and attenuation); I do not discuss the others. I do examples for parity and LRC and discuss the idea behind CRC. I do several examples of error correction by retransmission. I build a simple diagram and explain the roles of ACKs and NAKs. I then ask what happens if a message is lost (and let them discover the need for timeouts). I ask what happens if an ACK is lost (and let them discover the need for buffers and message numbers). I ask what happens if a NAK is lost (and let them discover that we really don't need NAKs, they just make the network faster).

I discuss each of the packets for the various data link layer protocols. My goal is to drive home the point that there is such a thing as packets at each level, that each protocol is different, and that many protocols are similar in structure (how they mark the start and end of a packet, where error control information and addresses are). I integrate the discussion of efficiency into each protocol by calculating its efficiency and asking the class how to improve it. I also introduce the idea that too large packets are a bad idea by asking why we don't have XMODEM-1meg. It is intuitive that this would be a bad idea and helps students understand why large packets aren't always the best.

Vivid examples help students understand media access control. We talk about conversation at social gatherings or parties as contention-based and discuss how people handle situations where more than one person begins to talk at the same time (like Ethernet). We talk about events where parliamentary procedure is used ("You now have the floor!") as using controlled access, like hub polling or Token Ring. In this way students realize they understood these concepts long before enrolling in a data communications course! Students are encouraged to observe "CSMA/CD" at parties (now they can fulfill a data communications assignment and go to a party at the same time!) and controlled access in Congress on C-SPAN.

War Stories

Squirrels and Impulse Noise

(Objective: illustrate the difficulties caused by impulse noise)

Recently our campus radio station received FCC approval to broadcast using a stronger signal. Immediately after they started broadcasting, the campus backbone network became unusable. It was filled with impulse noise. It took 2 days to link the impulse noise to the radio station, and when the radio station returned to its usual broadcast signal the problem disappeared.

However, this was only the first step in the problem. The radio station wanted to broadcast at full strength and there was no good reason for the stronger broadcast to affect the backbone in this way. After two weeks of effort, the problem was discovered. A short section of the backbone ran above ground between two buildings. It turns out that this specific brand of outdoor cable we used was particularly tasty to squirrels. They had eaten the outer insulating coating off of the cable, making it act like an antennae to receive the radio signals. The cable was replaced with a steel-coated armored cable and things worked fine when the radio station returned to its stronger signal.

Shielding Against Impulse Noise

(Objective: another example of impulse noise problems)

When I was an undergraduate, I worked during the summers in Halifax Nova Scotia for Sperry Univac (a then-important computer manufacturer, now Unisys). One of our clients was Power Corp, a large electrical utility with a computer room on the 7th floor of a downtown skyscraper. Every second Friday at about 4:00pm the computer network would suffer a short-duration (2-5 seconds) impulse noise hit that caused considerable problems.

It took a long time to identify that the noise was occurring at regular intervals, and even longer to identify the source. Halifax is the major east coast base of the Royal Canadian Navy, whose ships at the time were on a two-week rotation ending every second Friday. Just as they rounded the point into the harbor, the lead ship would blip its high power radar into the harbor and directly into our building, causing the network outage.

Obviously, asking the navy to stop was not a viable option. Instead, we coated that side of the computer room wall with aluminum foil to act as a shield. It worked fine, but looked rather silly. I have learned from one of my students that there is a company that now sells this type of insulating material for exactly this type of problem, and that the panels look like ordinary cubicle walls, not something out of the Apollo missions.